Assignment

SOCCER GAMES

- Certain editors at Sage like to think they are a bit of a whiz at soccer. To see whether they are better than Sussex lecturers and postgraduates we invited various employees of Sage to join in our soccer matches. Every player was allowed to play in only one match. Over many matches, we counted the number of players who scored goals.

Load:
SageEditorsPlaySoccer.dat

- Perform a chi-square test to see whether more publishers or academic scored goals. We predict that Sussex people will score more than Sage people.

Assignment

ONE-HANDED CYCLISTS

- There are cultural differences between Holland and England. The Dutch are famous for traveling by bike; they do it much more than the English. However, I noticed that many more Dutch people cycle while steering with only one hand. I pointed this out to one of my fiends, and she said that I was being a crazy English fool and that Dutch people did not cycle one-handed. Several weeks of me pointing at one-handed cyclists and her pointing at two-handed cyclists ensued. To put it to the test I counted the number of Dutch and English cyclists who ride with one or two hands on the handlebars.

Load:
Handlebars.dat

- Can you work on which one of us is right?

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Assignment

DARK LOAD

- There has been speculation about the influence of subliminal messages on records. A psychologist was interested in whether backward masked messages really influence people into doing strange behaviors. He took the master tapes of Britney Spears' song and created a modified version that had the backward message "deliver your soul to the dark lord" repeated in the chorus. He took both the original and modified versions and played one version (randomly) to a group of 32 butchers. He took the same group six months later and played them whatever version they hadn't heard the time before. So each person heard both the original and the modified version, but at different times. They psychologist measured the number of goats that were sacrificed in the week after listening to each version.

Load:
DarkLord.dat

- What is the hypothesis to be tested?
- Analyse the data with a Wilcoxon signed-rank test and report the effect size.
- Would the results be different using t-test?

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Assignment

CHECK THE PAPER

Biological Psychology 169 (2022) 108277

Contents lists available at ScienceDirect

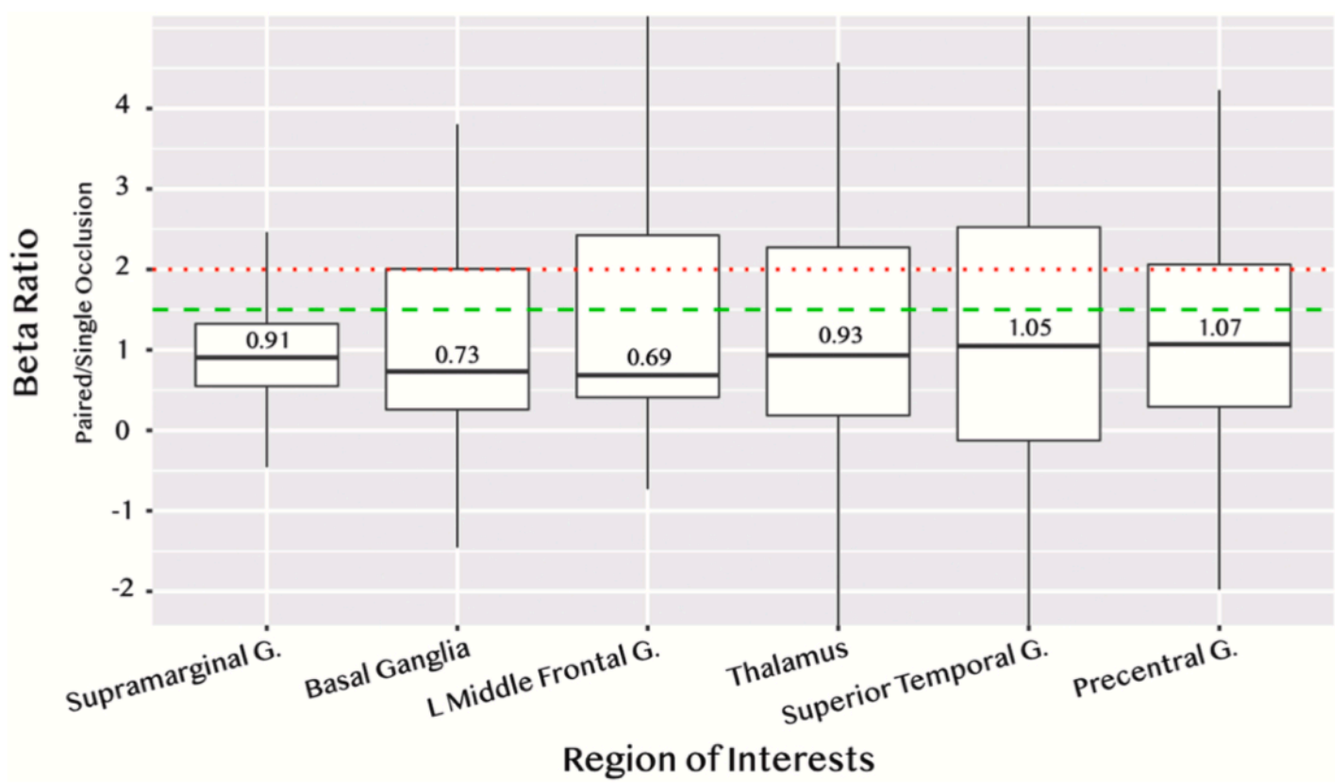
Biological Psychology

journal homepage: www.elsevier.com/locate/biopsycho

Load:
Ratio_Sub_by_Region.csv

Neural substrates of respiratory sensory gating: A human fMRI study

Pei-Ying S. Chan [a,b,*], Chia-Hsiung Cheng [a,b,c], Yu-Ting Wu [a,b], Changwei W. Wu [d,e,**], Ai-Ling Hsu [f], Chia-Yih Liu [b], Ho-Ling Liu [g], Paul W. Davenport [h]



- **Check the Research Report**
- **Please identify the statements of research question, hypothesis, and statistic testing methods.**
- **Please reproduce the results using the attached dataset in R. Report your own decision following the testing.**